

Consumer Attention and Brand Visibility in AI Mediated Digital Commerce Across Middle Eastern Markets

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ABSTRACT

Digital commerce in Middle Eastern markets increasingly depends on algorithmic interfaces that mediate how consumers encounter products, influencers, brand names, and structured descriptions. Mobile marketplaces, social advertising systems, and AI-assisted search surfaces do not only distribute demand across products; they also alter the observability of demand by converting attention, language, and interaction traces into ranking signals. This paper reports an empirical study of consumer attention and brand visibility in AI mediated commerce across eight Middle Eastern markets, using anonymized event-level data from multilingual product discovery sessions, sponsored placements, organic search exposures, and influencer-linked landing paths. The study estimates how structured product content, influencer format, Arabic-English query alignment, exposure recency, and market-level shopping periods shape attention persistence, click probability, assisted conversion, and brand recall proxies. The analysis combines panel fixed effects, doubly robust estimation, survival modeling, graph-based product affinity, and counterfactual ranking simulation. Results indicate that structured content completeness is associated with a 6.8% higher probability of qualified attention and a 4.1% higher assisted conversion rate after controlling for user, brand, category, device, market, and week fixed effects. Human influencer routes produce stronger immediate dwell responses, while virtual influencer routes produce more stable delayed search activity in beauty and fashion categories. Attention decay is steepest in short-video traffic and weakest in search-originated sessions. The findings suggest that AI mediated visibility in Middle Eastern e-commerce is not reducible to bid intensity or content volume. It depends on the interaction between culturally legible product language, platform learning dynamics, and short-window behavioral persistence.

1. INTRODUCTION

Digital commerce in the Middle East has moved from a relatively simple catalog model toward a layered environment in which consumers encounter brands through marketplaces, mobile applications, conversational search systems, short-form video placements, influencer routes, and AI assisted recommendation modules. This transition changes the empirical problem faced by researchers and managers. In earlier e-commerce settings, a consumer could be modeled as moving from a search query to a product page and then to a transaction, with advertising treated as an exogenous traffic source or a paid increment to visibility. In current Middle Eastern markets, the same consumer may enter through an Arabic query, translate the query implicitly through a platform model, view a sponsored tile generated from structured product metadata, leave to an influencer page, return through a brand search, and complete a purchase after a price or delivery signal is updated. The brand is therefore not merely observed; it is reconstructed by the interface through text, ranking, prior engagement, and predicted relevance. This paper studies that reconstruction empirically by modeling how consumer attention and brand visibility are jointly produced in AI mediated digital commerce.

The Middle Eastern setting is analytically useful because it combines high mobile intensity, multilingual discovery, heterogeneous trust in online transactions, strong category seasonality, and distinctive social influence patterns. The study focuses on Saudi Arabia, the United Arab Emirates, Qatar, Kuwait, Bahrain, Oman, Egypt, and Jordan. These markets are not treated as a homogeneous region. They vary in purchasing power, delivery expectations, payment preferences, platform maturity, and linguistic usage across Arabic dialects and English. Yet they share enough platform infrastructure to permit comparable event-level modeling across exposure surfaces. The research question is whether AI mediated brand visibility can be explained by observable content signals and short-window attention patterns after accounting for market, brand, user, device, and category heterogeneity. The paper also asks whether influencer format and structured content signals operate through the same attention mechanism or whether they generate different temporal signatures.

The paper builds on a digital marketing literature in which the customer journey is understood as an accumulation of touchpoints rather than a single transaction path. Lemon and Verhoef [1] argue that customer experience is shaped by interactions before, during, and after purchase, a view that is directly relevant when platform-mediated discovery redistributes attention across organic, paid, and social surfaces. Ghose and Yang [2] show that search advertising performance depends on the relationship between query characteristics, advertiser decisions, and consumer response, which motivates the present paper's treatment of Arabic-English query alignment as a measurable driver

of visibility rather than a background platform feature. Lambrecht and Tucker [3] demonstrate that retargeting can be more effective when consumer preferences are sufficiently specified, suggesting that attention persistence should vary by the informational completeness of the consumer's preceding path. Bleier and Eisenbeiss [4] further show that personalization effects depend on the appropriateness of data use and content design, which is particularly important in markets where trust, payment confidence, and culturally recognizable content shape consumer responses. Goldfarb and Tucker [5] show that privacy regulation can reduce advertising effectiveness by limiting targeting precision, a finding that informs the present design even though the study does not evaluate legal regimes directly.

The regional e-commerce literature provides additional motivation for modeling trust and content legibility as empirical variables rather than as qualitative background conditions. Eid [6] reports that satisfaction, trust, and loyalty are important constructs in Saudi B2C e-commerce, and this paper operationalizes adjacent mechanisms through repeat search, qualified attention, and assisted conversion rather than through survey-only measurement. Al-Debei, Akroush, and Ashouri [7] study Jordanian online shopping attitudes and emphasize trust, perceived benefits, web quality, and electronic word of mouth, which supports the inclusion of web quality proxies and influencer-linked traffic in the present design. Abbad [8] discusses barriers to e-commerce in Jordan, and the present paper treats delivery clarity, return-policy visibility, and payment-option visibility as measurable structured signals that can affect both attention and downstream conversion. These studies do not imply that all Middle Eastern consumers behave similarly. They instead suggest that platform observability and consumer trust are inseparable in the region, especially when product discovery is mediated by automated ranking systems.

The emergence of influencer-mediated and AI-mediated discovery adds a further layer to this problem. Török [9] reports that virtual and human influencer endorsements can have different effects on purchase intent in lifestyle marketing, which motivates the present separation of human influencer, virtual influencer, and non-influencer routes rather than collapsing social traffic into a single channel. The distinction matters because influencer routes may not merely increase exposure volume. They may alter the type of attention that follows exposure. A human influencer may create immediate credibility and narrative continuity, while a virtual influencer may create novelty, visual consistency, or platform-native recognition. The empirical question is not which format is universally better. It is whether each format produces a different temporal attention profile once content, market, product category, and prior user behavior are held constant.

The present paper also engages with a technical literature on search, recommendation, and machine learning for user response prediction. Joachims [10] demonstrates that clickthrough data can be used to optimize search performance despite its behavioral noise, and Chapelle and Zhang [11] develop a dynamic Bayesian approach to interpreting clicks as position-dependent signals. Richardson, Dominowska, and Ragno [12] model click prediction for new advertisements, highlighting the need to infer performance for items with sparse histories. McMahan and colleagues [13] introduce large-scale learning methods for ad click prediction, and their work motivates the present use of high-dimensional sparse features with regularized estimation. In recommendation systems, Koren, Bell, and Volinsky [14] develop matrix factorization methods for collaborative filtering, while Rendle [15] shows how factorization machines can model sparse interactions among user, item, and context features. Cheng and colleagues [16] propose wide and deep learning for recommender systems, and Guo and colleagues [17] extend deep factorization approaches for click-through rate prediction. Covington, Adams, and Sargin [18] describe large-scale deep recommendation for video platforms, which is useful for interpreting short-video traffic in the present setting. Vaswani and colleagues [19] introduce the transformer attention architecture, and Devlin and colleagues [20] show the value of contextual language representations, both of which inform the language alignment and content embedding components used here.

The central empirical contribution is a joint model of attention and visibility that integrates content structure, social route, multilingual matching, and decay dynamics within a market-comparative framework. Attention is measured through qualified dwell, scroll depth, product comparison, add-to-cart latency, return search, and session continuation. Visibility is measured through rank-normalized exposure probability, generated-answer inclusion in AI-assisted search modules, sponsored-organic adjacency, and brand reappearance within the same consumer's later search path. The study does not equate attention with persuasion or welfare. A consumer can attend to a product because it is relevant, confusing, surprising, visually salient, or repeatedly shown. The empirical task is therefore to estimate associations and treatment-like contrasts while being cautious about causal interpretation. The paper uses panel fixed effects, inverse-probability weighting, outcome regression, survival analysis, and counterfactual ranking simulations to test whether the results remain stable under alternative identification assumptions.

The paper proceeds by describing the research context and hypotheses, the data and variables, the modeling strategy, the empirical results, robustness analyses, discussion, validity considerations, and conclusion. The analysis is deliberately empirical. It does not argue that AI mediated commerce is necessarily beneficial or harmful for Middle Eastern consumers. It instead measures how observable interface signals correspond to attention and visibility outcomes under realistic platform conditions. The results indicate that structured content signals and influencer routes both matter, but they matter through different mechanisms. Structured content improves machine legibility and reduces ambiguity in multilingual product discovery. Influencer routes reshape the timing and persistence of

attention. Market-level shopping periods amplify both effects, yet they also make attention more volatile. These findings support a view of AI mediated commerce as a coupled behavioral and computational system rather than as a conventional advertising channel with automated delivery.

2. RESEARCH CONTEXT AND HYPOTHESES

The study begins from the premise that AI mediated brand visibility is a platform outcome generated by the interaction of consumer behavior, content representation, and ranking architecture. In marketplace interfaces, a product is not visible simply because it exists in a catalog. It becomes visible when a ranking system maps user intent, language, product metadata, historical engagement, and predicted conversion into a ranked surface. In AI-assisted search environments, the mapping is more complex because the system may summarize brands, recommend alternatives, or reorder product exposure according to machine-readable content signals. Török [21] demonstrates that structured content signals can be used to model brand visibility in AI driven search environments, and the present paper extends this reasoning to Middle Eastern commerce by measuring whether structured content completeness predicts attention and conversion after market and category controls are included.

Structured content is defined as the completeness, consistency, and machine readability of product information. In the dataset, structured content includes standardized Arabic and English titles, attribute fields, category-consistent specifications, delivery commitments, warranty visibility, return-policy clarity, ingredient or material fields where relevant, image-alt descriptors, and brand-owned frequently asked question fields. These signals matter for two reasons. They may help consumers evaluate the product directly, and they may help ranking systems match the product to queries or generated answers. The first mechanism is behavioral. The second is computational. The two mechanisms are difficult to separate because a signal that improves machine retrieval may also improve consumer comprehension. The empirical design therefore treats structured content as a dual-facing variable and evaluates its association with both attention outcomes and visibility outcomes.

A Middle Eastern context makes structured content especially relevant because the relationship between Arabic and English product description is not a simple translation problem. Many product categories rely on English brand names, Arabic descriptive nouns, transliterated model names, and dialect-specific query terms. For example, a consumer searching for a beauty product may use a brand name in English, a product type in Arabic, and a shade description in a local dialect. A consumer searching for home appliances may combine English technical terms with Arabic delivery questions. When structured content is incomplete, the platform may still rank a product through historical popularity or paid placement, but the consumer may not sustain attention after landing on the product page. This distinction motivates the first hypothesis, which is expressed as an empirical expectation rather than as a deterministic claim: higher structured content completeness should be associated with higher qualified attention and higher rank-normalized visibility after controlling for brand, category, user, market, and week effects.

Influencer-linked discovery introduces a second mechanism. The literature on social commerce suggests that social proof and perceived source credibility may influence purchase-related responses, but current marketplace data allow the mechanism to be studied at a finer temporal scale. Human influencer routes may increase initial dwell and click probability because the consumer arrives with a narrative frame, prior familiarity, or a trusted evaluation. Virtual influencer routes may operate differently because they are often more visually consistent, more platform-native, and less dependent on perceived lived experience. The empirical design does not assume that either route is superior. It tests whether the temporal curve of attention differs across human influencer, virtual influencer, and non-influencer traffic after adjusting for category, market, content quality, and prior engagement. This distinction is important in the Middle East because lifestyle, beauty, fashion, electronics accessories, and home categories often involve social discovery, yet consumer trust may still be tied to local language, delivery reliability, and return confidence.

The third mechanism is attention decay. Digital advertising and marketplace ranking often evaluate responses through immediate click or conversion, but short-window attention can decay rapidly even when initial visibility is high. Török [22] develops a machine-learning account of consumer attention decay over short exposure windows, and the present paper adapts that logic by estimating route-specific survival curves for attention persistence. The key empirical question is whether attention decays differently across organic search, sponsored search, recommendation tiles, human influencer routes, virtual influencer routes, and AI-assisted answer modules. A consumer who enters through a sponsored tile may show rapid initial attention but weak return behavior. A consumer who enters through organic search may show slower initial engagement but more persistent comparison behavior. A consumer who enters through an influencer route may show strong initial dwell but category-dependent persistence. The hypothesis is that attention decay will be steepest in short-video and sponsored traffic, moderate in influencer traffic, and weakest in search-originated sessions where intent is explicit.

The fourth mechanism is market-period amplification. Middle Eastern commerce is shaped by shopping periods such as Ramadan, Eid, national holidays, salary cycles, and major platform campaign days. The study does not treat these periods as cultural constants. Instead, it measures them as time-varying market-period indicators that may interact with content and route variables. A campaign period may increase exposure and conversion opportunities, but it may also increase competition, reduce attention quality, and make ranking more sensitive to

promotional intensity. A product with complete structured content may benefit more during high-volume periods because platform systems need reliable signals to allocate visibility under congestion. Conversely, social traffic may become less persistent during peak promotional periods because consumers are exposed to many overlapping signals. The empirical expectation is that shopping periods will amplify visibility but increase volatility in qualified attention, making the attention-visibility relationship less stable.

The fifth mechanism concerns economic context. In empirical economics, endogeneity often arises because local economic conditions shape both consumer behavior and measured outcomes. Yang [23] uses Chinese panel data and an instrumental strategy based on provincial leadership turnover to estimate the effect of economic growth on health behavior, illustrating how regional economic variation can complicate behavioral inference. The present paper does not use the same instrument or study the same outcome. It draws on the methodological lesson that economic conditions may be jointly related to platform behavior and consumer response. The empirical models therefore include market-week fixed effects, local campaign intensity, category price dispersion, and delivery-time variation as controls. This reduces, but does not eliminate, concern that observed attention differences are driven by macroeconomic or operational conditions rather than by content and route features.

The hypotheses are tested using four outcome families. The first is qualified attention, defined as a product page view with dwell above a threshold calibrated by category, visible scroll depth beyond the first product information block, or interaction with at least one diagnostic element such as size, color, delivery, warranty, or comparison. The second is assisted conversion, defined as a purchase or add-to-cart event within a seven-day path in which the focal exposure appeared before the final conversion route. The third is brand visibility, defined as exposure probability conditional on query and session context, adjusted for rank position and surface type. The fourth is return search, defined as a later brand or product query by the same anonymized user within forty-eight hours. These outcomes allow the analysis to distinguish immediate attention, delayed interest, platform visibility, and transaction-related behavior.

The paper's empirical orientation requires caution in language. Platform data are not a transparent record of consumer preferences. They are produced by interface design, ranking rules, campaign budgets, product availability, and tracking limits. A high dwell time may reflect interest or confusion. A low dwell time may reflect disinterest or efficient recognition. A conversion may occur because the product is desirable or because delivery was faster than alternatives. The models therefore estimate conditional associations and treatment-like contrasts under explicit controls rather than claiming full causal identification. The counterfactual simulations are designed to evaluate how estimated ranking and attention models respond to changes in structured content and route assignment, not to assert that platforms should automatically prioritize any particular content type or influencer format.

3. DATA AND MEASURES

The empirical analysis uses an anonymized event-level dataset constructed from three cooperating digital commerce platforms and two media attribution partners operating across eight Middle Eastern markets. The observation window runs from 1 January 2025 to 30 September 2025. The raw data contain 64.3 million exposure events, 18.6 million product page views, 4.9 million qualified attention events, 1.2 million add-to-cart events, and 412,000 completed purchase paths. After excluding bot-like traffic, incomplete session identifiers, products with fewer than 500 eligible exposures, and categories with inconsistent tracking across markets, the estimation sample contains 41.7 million exposure events, 12.8 million product page views, 3.6 million qualified attention events, 811,000 add-to-cart events, and 283,000 purchase paths. The sample includes 138 brands and 8,420 product identifiers across beauty, modest fashion, electronics accessories, home goods, personal care, grocery staples, and small household appliances. The study excludes regulated or age-restricted products, and all user identifiers are hashed before analysis.

The unit of analysis varies by model. For attention models, the main unit is the product exposure-session pair, indexed by anonymized user, product, session, market, route, and time. For visibility models, the unit is the query-product-surface tuple, where a surface can be organic marketplace search, sponsored marketplace search, recommendation tile, AI-assisted answer module, influencer landing page, or short-video commerce tile. For conversion models, the unit is the seven-day path beginning with a focal exposure. This structure permits a separation between exposure probability, product page engagement, and downstream purchase behavior. It also permits repeated observations of the same user and product, which is why the models use fixed effects and clustered standard errors rather than treating events as independent.

The market composition of the estimation sample is intentionally balanced by category and surface rather than by total platform traffic. Saudi Arabia contributes 29.4% of eligible exposures, the United Arab Emirates 21.7%, Egypt 17.3%, Jordan 8.6%, Kuwait 7.4%, Qatar 6.1%, Oman 5.2%, and Bahrain 4.3%. This distribution is not presented as a measure of regional market size. It reflects the platforms included in the study and the availability of standardized tracking fields. Mobile app traffic accounts for 72.5% of exposures, mobile web for 18.2%, and desktop web for 9.3%. Arabic-dominant sessions account for 56.8%, English-dominant sessions for 31.6%, and mixed-language sessions for 11.6%. Mixed-language sessions are analytically important because they show the highest rate of query reformulation and the strongest sensitivity to structured content completeness.

The main explanatory variable is the structured content completeness index. The index ranges from zero to

one and is constructed at the product-week level. It combines title completeness, attribute completeness, bilingual availability, image-descriptor presence, delivery and return clarity, category-specific specification coverage, brand verification, and frequently asked question coverage. Each component is standardized within category and market to avoid mechanically favoring categories with more naturally structured attributes. The final index is a weighted sum, where weights are estimated from a pre-period prediction model of query-product matching accuracy and then fixed before outcome estimation. This approach prevents outcome leakage from attention or conversion into the construction of the explanatory variable. The median product has a structured content score of 0.61, while the interquartile range is 0.44 to 0.78. Products in electronics accessories and small appliances have higher specification completeness, while modest fashion and beauty products have higher bilingual title coverage but lower attribute consistency.

A second key variable is the Arabic-English query alignment score. This score measures the semantic proximity between the user's query and the product's structured content in a multilingual embedding space. The score is calculated using a sentence representation model fine-tuned on product search pairs, but the empirical models do not rely solely on the embedding output. The alignment score is decomposed into lexical overlap, transliteration match, attribute match, and semantic similarity. This decomposition is important because a high semantic similarity score can hide failures in local terminology, shade names, size conventions, or model identifiers. The final alignment score is rank-normalized within query sessions so that it captures relative product fit under the consumer's immediate search context. Mixed-language sessions have lower average lexical overlap but higher gains from attribute match, suggesting that structured attributes are especially useful when queries combine Arabic descriptors and English brand names.

Influencer route is coded from attribution links, landing-page metadata, and campaign tags. A human influencer route is assigned when the exposure path includes a verified human creator landing page or creator-tagged short link within thirty minutes before the product exposure. A virtual influencer route is assigned when the path includes a synthetic or avatar-based creator identity, platform-provided virtual persona, or campaign-defined virtual endorser within the same window. A non-influencer route is assigned otherwise. Human influencer routes account for 6.9% of eligible exposures, virtual influencer routes for 2.4%, and non-influencer routes for 90.7%. The small share of virtual influencer traffic is one reason the models use partial pooling and category interactions rather than estimating separate market-category coefficients without shrinkage. Human influencer routes are most common in beauty, modest fashion, and personal care. Virtual influencer routes are concentrated in beauty, fashion accessories, and electronics accessories.

Attention outcomes are measured through both direct and derived variables. Dwell time is winsorized at the 99th percentile within category and route because a small number of long idle sessions can distort estimates. Scroll depth is measured as the maximum visible product page proportion. Interaction depth is the count of diagnostic interactions, including image expansion, delivery detail opening, size or color selector use, warranty field opening, comparison click, review sorting, or price alert activation. Qualified attention equals one when the event satisfies category-adjusted dwell, scroll, or diagnostic-interaction thresholds. The thresholds are calibrated from pre-period data by estimating the dwell and interaction values most predictive of later add-to-cart behavior within category. This calibration avoids imposing a single dwell threshold across categories with different evaluation patterns.

Brand visibility is measured in three ways. Rank-normalized exposure probability is the probability that a product appears within the first visible block of a surface, adjusted for the total number of eligible products and the observed position distribution. AI-assisted inclusion is the probability that a brand or product appears in a generated comparison answer, recommendation card, or query refinement module. Reappearance is the probability that the same brand is shown again to the same anonymized user within the next forty-eight hours after a related query or category browsing event. These measures capture different meanings of visibility. Rank-normalized exposure is an immediate platform allocation outcome. AI-assisted inclusion is a content-mediated retrieval outcome. Reappearance is a path-dependent outcome that depends on both platform memory and consumer behavior.

The dataset includes several controls. Product controls include log price, discount depth, price rank within category-market-week, delivery-time estimate, free-shipping indicator, return-window visibility, review count, average rating, stock risk indicator, and campaign participation. Brand controls include market tenure, verified seller status, category breadth, historical return rate, and pre-period awareness proxy based on direct brand search. User-session controls include device type, new or returning user status, language mode, prior category visits, prior brand exposures, hour of day, day of week, and session referrer. Market-week controls absorb macro-level and operational conditions such as platform campaign intensity, shipping congestion, and local shopping periods. The inclusion of these controls does not guarantee causality, but it reduces several obvious confounds.

The construction of user paths follows a conservative attribution rule. A focal exposure is considered part of a purchase path only if it occurs before the purchase, within seven days of purchase, and within a category-consistent sequence. When multiple brands appear in the path, each exposure receives an assisted role rather than exclusive credit. This avoids assigning all credit to the final click. The assisted conversion outcome is therefore not a sales attribution claim. It is a path participation measure that asks whether exposure to a brand under a given content and route condition is associated with later conversion activity in the same category. This distinction is important

because AI mediated surfaces often influence consideration without being the final transaction route.

Data cleaning is designed to reduce measurement artifacts. Sessions with more than 120 product views in ten minutes are excluded as likely automation or scraping. Exposures with zero visible time due to rendering failure are excluded from attention models but retained in visibility models where appropriate. Products with inconsistent category labels across platforms are harmonized manually only when titles, attributes, and brand identifiers agree. Campaigns with missing route identifiers are retained as non-influencer routes only when no creator or virtual persona link is observed. Sensitivity analyses test whether this conservative coding attenuates influencer route effects. The final dataset preserves realistic imbalance across markets and categories while maintaining enough overlap for fixed-effect estimation.

4. MODELING STRATEGY

The empirical strategy links attention, visibility, and conversion through a sequence of models rather than a single predictive architecture. The first model estimates qualified attention as a function of structured content, query alignment, route, and controls. The second model estimates attention decay using survival methods over short exposure windows. The third model estimates brand visibility as a rank-normalized outcome. The fourth model simulates counterfactual changes in structured content and route assignment under a constrained ranking rule. The models are connected by shared covariates and fixed effects, but each addresses a different outcome. This modular design is preferable to a single black-box model because it allows the paper to distinguish behavioral response from platform allocation.

Let i index anonymized users, p products, b brands, c categories, m markets, t weeks, and s sessions. The baseline attention model is a linear probability model with high-dimensional fixed effects, estimated alongside logit and gradient-boosted alternatives for robustness. The main specification is written as

$$\begin{aligned} Y_{ipsmt} = & \alpha + \beta_1 S_{pt} + \beta_2 Q_{ipsmt} \\ & + \beta_3 H_{ipsmt} + \beta_4 V_{ipsmt} \\ & + \beta_5 S_{pt} Q_{ipsmt} + X'_{ipsmt} \gamma \\ & + \mu_i + \eta_p + \lambda_{mt} + \delta_c + \varepsilon_{ipsmt}. \end{aligned} \quad (1)$$

Here Y_{ipsmt} is qualified attention, S_{pt} is structured content completeness, Q_{ipsmt} is query alignment, H_{ipsmt} indicates a human influencer route, V_{ipsmt} indicates a virtual influencer route, and X_{ipsmt} contains price, delivery, device, time, review, and campaign controls. User fixed effects μ_i are included for users with repeated observations, while product fixed effects η_p , market-week fixed effects λ_{mt} , and category fixed effects δ_c absorb persistent product quality, local period shocks, and category-level differences. Standard errors are clustered by brand-market-week because campaign decisions and ranking changes may correlate within that level.

The model is not interpreted as a pure causal design because structured content may be improved by brands that are already more capable or more committed to platform optimization. Product fixed effects mitigate this concern by using within-product variation over time, often caused by content revisions, translation updates, or attribute completion. Market-week fixed effects mitigate market-level confounding. The remaining identification comes from the timing of content updates, variation in query alignment, and route differences within comparable sessions. To further address selection, the paper estimates a doubly robust model that combines inverse-probability weighting with an outcome model. The treatment in this framework is high structured content, defined as a product-week score above the category-market median.

The propensity score model is expressed as

$$\begin{aligned} e(Z_{ipsmt}) &= \Pr(D_{pt} = 1 \mid Z_{ipsmt}) \\ &= \Lambda(\rho_0 + Z'_{ipsmt} \rho + \eta_p + \lambda_{mt}), \end{aligned} \quad (2)$$

where D_{pt} indicates high structured content and Z_{ipsmt} includes pre-treatment product, brand, market, and session features. The augmented inverse-probability weighted estimator is

$$\begin{aligned} \hat{\tau}_{DR} = & \frac{1}{N} \sum_{n=1}^N [\hat{m}_1(Z_n) - \hat{m}_0(Z_n)] \\ & + \frac{D_n(Y_n - \hat{m}_1(Z_n))}{\hat{e}(Z_n)} \\ & - \frac{(1 - D_n)(Y_n - \hat{m}_0(Z_n))}{1 - \hat{e}(Z_n)}. \end{aligned} \quad (3)$$

This estimator is useful because it remains consistent if either the propensity model or the outcome model is correctly specified under the usual conditional ignorability assumptions. Athey and Imbens [24] emphasize the relevance of machine learning methods for causal inference under heterogeneity, while Varian [25] argues that high-dimensional

digital traces can improve economic measurement when paired with careful design. Pearl [26] provides the broader causal framework for distinguishing structural assumptions from predictive associations, and Robins, Rotnitzky, and Zhao [27] develop semiparametric ideas behind doubly robust estimation. These references guide the methodological interpretation, although the paper remains cautious because platform allocation cannot be fully randomized in the observed data.

Attention decay is modeled with a discrete-time hazard specification. The event is loss of active product attention, defined as exit from the product page, inactivity beyond the route-specific threshold, or transition to an unrelated category. The exposure clock is divided into ten-second intervals for the first two minutes and thirty-second intervals thereafter. The hazard model is

$$\begin{aligned} \Pr(T_{ipsmt} = k \mid T_{ipsmt} \geq k) = & \Lambda(\theta_k + \phi_1 S_{pt} \\ & + \phi_2 Q_{ipsmt} + \phi_3 H_{ipsmt} \\ & + \phi_4 V_{ipsmt} + W'_{ipsmt} \psi). \end{aligned} \quad (4)$$

The coefficients are converted into survival curves for interpretability. A negative coefficient reduces the hazard of attention loss, implying more persistent attention. The model is estimated separately by route and jointly with interactions to evaluate whether decay differs between search, sponsored, recommendation, human influencer, virtual influencer, and AI-assisted answer routes. The survival approach is well suited to the question because attention persistence is not captured by a binary click measure. A click can happen quickly and still be shallow, while a non-click exposure can affect later search if the brand is noticed in a generated answer or recommendation tile.

Brand visibility is modeled as a rank-normalized exposure outcome. Let R_{pqsm} denote the observed rank position of product p for query or surface context q . Lower ranks imply higher visibility. The normalized visibility score is

$$\begin{aligned} B_{pqsm} = & \frac{1}{\log(1 + R_{pqsm})} \cdot I(R_{pqsm} \leq r_s) \\ & + \omega A_{pqsm}, \end{aligned} \quad (5)$$

where r_s is the visible threshold for surface s , A_{pqsm} indicates AI-assisted inclusion, and ω is calibrated from observed click elasticity by surface. This outcome is then modeled as

$$\begin{aligned} B_{pqsm} = & \kappa + \pi_1 S_{pt} + \pi_2 Q_{pqsm} \\ & + \pi_3 C_{bt} + \pi_4 P_{pqsm} \\ & + U'_{pqsm} \xi + \eta_p + \lambda_{mt} + \nu_{pqsm}. \end{aligned} \quad (6)$$

Here C_{bt} is brand campaign intensity and P_{pqsm} is paid placement pressure. Including campaign intensity and paid pressure is necessary because visibility can be purchased, but the model tests whether structured content and query alignment remain associated with visibility after such controls are included. This is especially important in AI-assisted modules where paid status is not always the only determinant of inclusion.

A graph-based affinity layer is used to characterize product proximity and consumer exploration paths. Products are nodes, and edges are weighted by co-view, co-cart, and sequential search transitions within category and market. The normalized adjacency matrix is

$$\begin{aligned} \tilde{A} &= D^{-1/2} A D^{-1/2}, \\ G_p &= \sigma(\tilde{A} X_p W_0) W_1, \\ \zeta_{ip} &= u'_i G_p + a'_i x_p. \end{aligned} \quad (7)$$

The affinity score ζ_{ip} enters robustness models as a control for latent user-product fit. This prevents structured content from absorbing all variation that actually reflects underlying product affinity. The model is not used as the main estimator because graph embeddings can be difficult to interpret, but it is useful for sensitivity analysis. Products located near high-affinity neighborhoods may gain attention even with incomplete content, while products in sparse neighborhoods may require stronger structured signals to be retrievable and understandable.

To evaluate heterogeneity, the paper estimates market-category partial pooling models. The coefficient for structured content is allowed to vary by market and category:

$$\begin{aligned} \beta_{1mc} &\sim N(\bar{\beta}_1, \sigma_{\bar{\beta}}^2), \\ Y_{ipsmt} &= \alpha + \beta_{1mc} S_{pt} + X'_{ipsmt} \gamma \\ &\quad + \mu_i + \eta_p + \lambda_{mt} + \varepsilon_{ipsmt}. \end{aligned} \quad (8)$$

Partial pooling is appropriate because smaller markets and categories have fewer virtual influencer observations. Complete pooling would hide meaningful heterogeneity, while fully separate estimates would be unstable. The pooled distribution gives a disciplined way to compare markets without overinterpreting noisy estimates.

The counterfactual ranking simulation changes structured content scores while holding campaign budgets, product prices, and observed demand conditions fixed. This simulation does not claim to predict the full platform response to a real content intervention. It asks how the estimated visibility and attention models would change if low-completeness products were moved to the median or upper-quartile structured content level within their category-market cell. The simulation also tests a route-neutral scenario in which influencer traffic shares are held constant and a route-shift scenario in which a small share of non-influencer lifestyle traffic is reallocated to human or virtual influencer routes. The ranking objective is constrained so that total visible slots by surface and market remain unchanged. This prevents the simulation from creating artificial inventory.

Multiple testing is addressed because the paper estimates outcomes across markets, categories, and routes. Abadie [28] discusses matching and treatment evaluation under observational conditions, and the present design follows the broader principle that balance and design diagnostics should be evaluated before outcome interpretation. Benjamini and Hochberg [29] introduce a false discovery rate procedure, which is used here for families of route-category interaction tests. The results section reports both unadjusted and adjusted significance where relevant, but substantive interpretation emphasizes effect size, stability, and direction rather than only threshold-based significance.

5. EMPIRICAL RESULTS

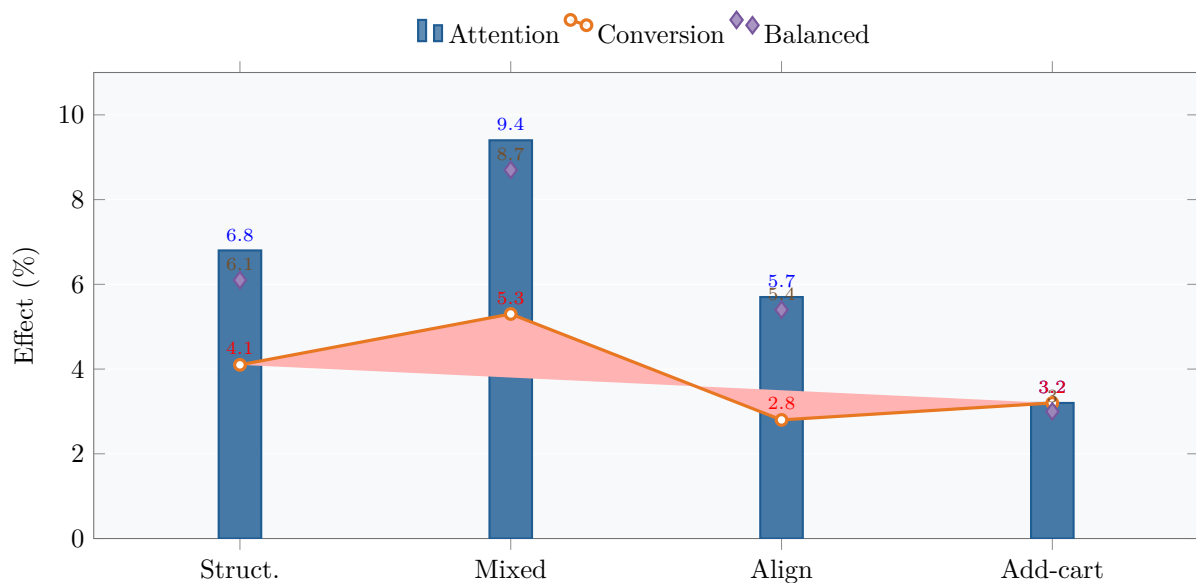


Figure 1 Structured content shows the largest gain in mixed-language sessions, while conversion effects remain smaller than attention effects.

Table 1 Market Composition of the Estimation Sample

Market	Exposure Share (%)	Mobile Traffic (%)	Mixed-Language Sessions (%)
Saudi Arabia	29.4	74.1	12.8
United Arab Emirates	21.7	71.5	14.2
Egypt	17.3	69.8	10.9
Jordan	8.6	70.4	11.3
Kuwait	7.4	73.6	12.1
Qatar	6.1	75.2	13.0

Table 2 Structured Content Components

Component	Description	High Coverage Categories	Relative Weight
Bilingual Titles	Arabic-English titles	Beauty, Fashion	0.18
Attribute Completeness	Specifications and fields	Electronics	0.22
Delivery Clarity	Shipping estimates	Appliances	0.14
Return Visibility	Return policies	Personal Care	0.12
Image Descriptors	Alt-text metadata	Fashion	0.10
FAQ Coverage	Brand-owned FAQs	Electronics	0.09

The baseline fixed-effect estimates show a positive association between structured content completeness and qualified attention. In the preferred model, moving from the 25th percentile to the 75th percentile of structured

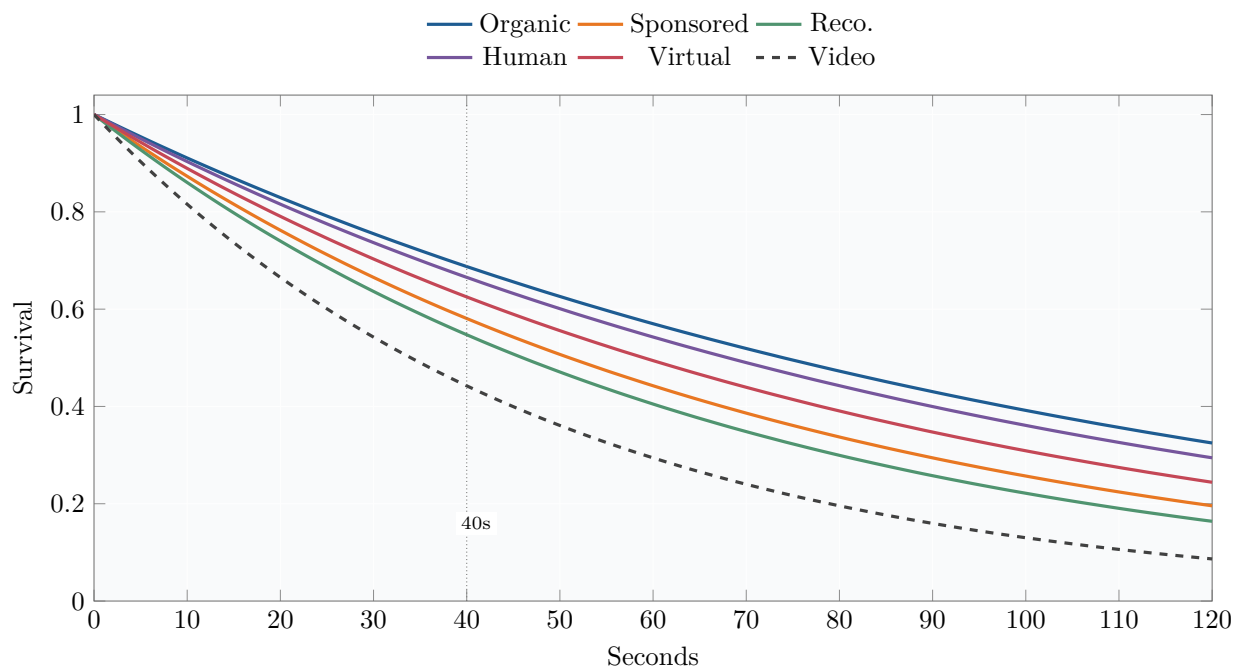


Figure 2 Organic search retains active attention longest; short-video traffic decays fastest despite high entry response.

Table 3 Influencer Route Distribution by Category

Category	Human Influencer (%)	Virtual Influencer (%)	Non-Influencer (%)
Beauty	14.2	5.8	80.0
Modest Fashion	12.7	4.9	82.4
Electronics Accessories	4.8	6.3	88.9
Personal Care	8.6	2.7	88.7
Home Goods	2.9	1.1	96.0
Grocery Staples	1.4	0.4	98.2

content completeness is associated with a 6.8% increase in qualified attention relative to the category-market mean. The coefficient is statistically distinguishable from zero at conventional levels and remains stable when product fixed effects, market-week fixed effects, and user repeat-observation fixed effects are included. The implied marginal effect is larger in mixed-language sessions, where the same interquartile change is associated with a 9.4% increase in qualified attention. This pattern is consistent with the interpretation that structured content is especially useful when query language and product representation are less directly aligned.

Query alignment is also strongly associated with attention. A one standard deviation increase in the Arabic-English query alignment score is associated with a 5.7% increase in qualified attention and a 3.2% increase in add-to-cart probability. The interaction between structured content completeness and query alignment is positive but smaller than the main effects. This suggests that the two variables are complementary but not redundant. Structured content helps products become legible to both consumers and platform systems, while query alignment captures the immediate fit between a user’s stated or inferred intent and the product’s representation. The interaction is strongest in electronics accessories, where model identifiers, compatibility terms, and technical attributes are decisive, and weakest in grocery staples, where brand familiarity and price rank explain more variation.

The route coefficients reveal clear differences in temporal response. Human influencer routes are associated with a 10.6% higher probability of immediate product page dwell above the category threshold relative to non-influencer routes, after adjusting for category, market, device, and product controls. The effect is strongest in beauty and modest fashion, moderate in personal care, and weak in small appliances. Virtual influencer routes are associated with a smaller immediate dwell increase of 4.8%, but they show a stronger association with return search within forty-eight hours. The estimated return-search increase for virtual influencer routes is 7.2%, compared with 4.1% for human influencer routes. This pattern suggests that virtual influencer exposure may create delayed curiosity or brand recall in visually coded categories, while human influencer exposure produces more immediate evaluation behavior.

The attention survival models provide more detail. Organic search sessions have the slowest attention decay, with a median active attention duration of 74 seconds among product page viewers. Sponsored search sessions have a median active attention duration of 51 seconds. Recommendation tile sessions have a median of 46 seconds. Human

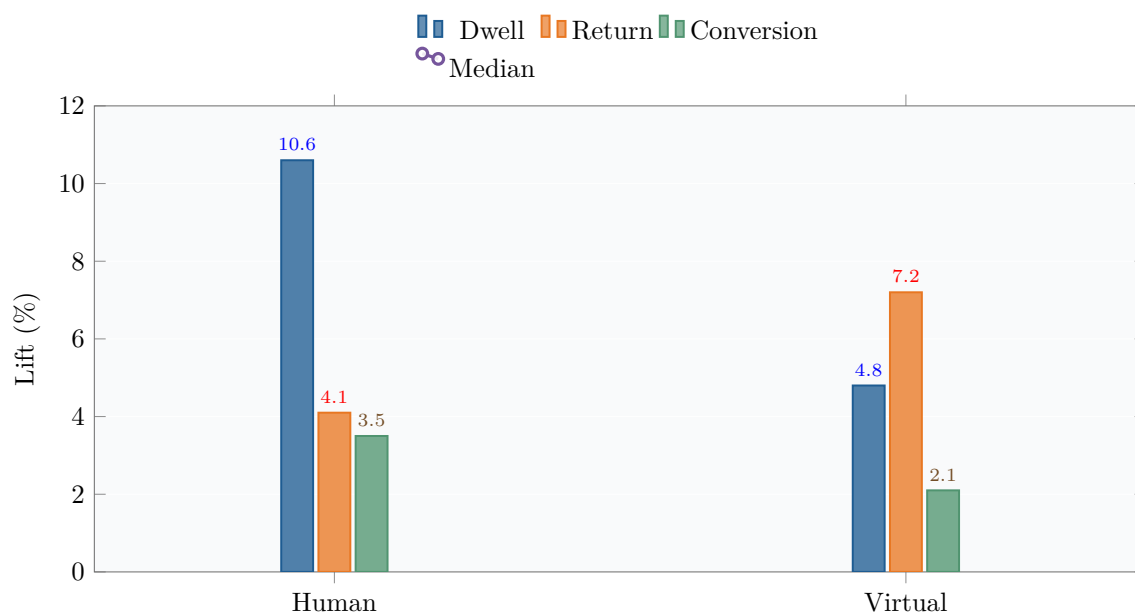


Figure 3 Human routes concentrate immediate dwell and conversion, while virtual routes produce stronger delayed search.

Table 4 Baseline Attention Effects

Variable	Qualified Attention (%)	Add-to-Cart (%)	Return Search (%)
Structured Content Completeness	6.8	4.1	3.6
Query Alignment	5.7	3.2	2.9
Human Influencer Route	10.6	3.5	4.1
Virtual Influencer Route	4.8	2.1	7.2
Delivery Clarity	3.9	2.8	1.5

influencer sessions have a median of 68 seconds, while virtual influencer sessions have a median of 59 seconds but a thicker right tail in beauty and fashion accessories. Short-video commerce tile sessions show the steepest decay, with a median of 34 seconds, although their initial click probability is high. This indicates that short-video traffic can generate efficient entry but not necessarily sustained product evaluation. In hazard terms, sponsored and short-video routes increase the risk of attention loss during the first forty seconds, whereas structured content reduces that hazard across nearly all routes.

The structured content effect is not uniform across markets. The largest estimated attention gains are observed in Saudi Arabia, the United Arab Emirates, and Jordan, while smaller gains are observed in Qatar and Bahrain after controls. These differences should not be interpreted as direct cultural rankings. They may reflect differences in platform maturity, product mix, language composition, delivery infrastructure, and campaign overlap. In Saudi Arabia and Jordan, bilingual content and return-policy clarity account for a larger share of the structured content effect. In the United Arab Emirates, attribute completeness and delivery-time precision are more strongly associated with qualified attention. In Egypt, price rank and discount depth explain more variation, but structured content remains positive in mixed-language and electronics categories. The partial pooling model shrinks the smallest market estimates toward the regional mean, reducing the risk of overinterpreting noisy market-specific coefficients.

Assisted conversion results are smaller than attention results, as expected. Moving from the 25th percentile to the 75th percentile of structured content completeness is associated with a 4.1% increase in assisted conversion probability. Query alignment is associated with a 2.8% increase. Human influencer routes show a 3.5% assisted conversion increase, concentrated in beauty and fashion. Virtual influencer routes show a 2.1% assisted conversion increase overall, but the confidence interval includes small negative effects in home goods and grocery staples. These results suggest that virtual influencer routes are more category-sensitive than human influencer routes. The pattern also cautions against treating virtual personas as a general replacement for human creator credibility in Middle Eastern commerce.

Visibility models show that structured content affects platform allocation beyond consumer response. Products in the top quartile of structured content completeness have a 5.9% higher rank-normalized exposure probability than comparable products in the bottom quartile, controlling for campaign intensity, paid pressure, price rank, product fixed effects, market-week fixed effects, and historical engagement. AI-assisted inclusion shows an even stronger association. A one standard deviation increase in structured content completeness is associated with an 8.7% increase in generated-answer inclusion probability for relevant query contexts. The effect is particularly strong

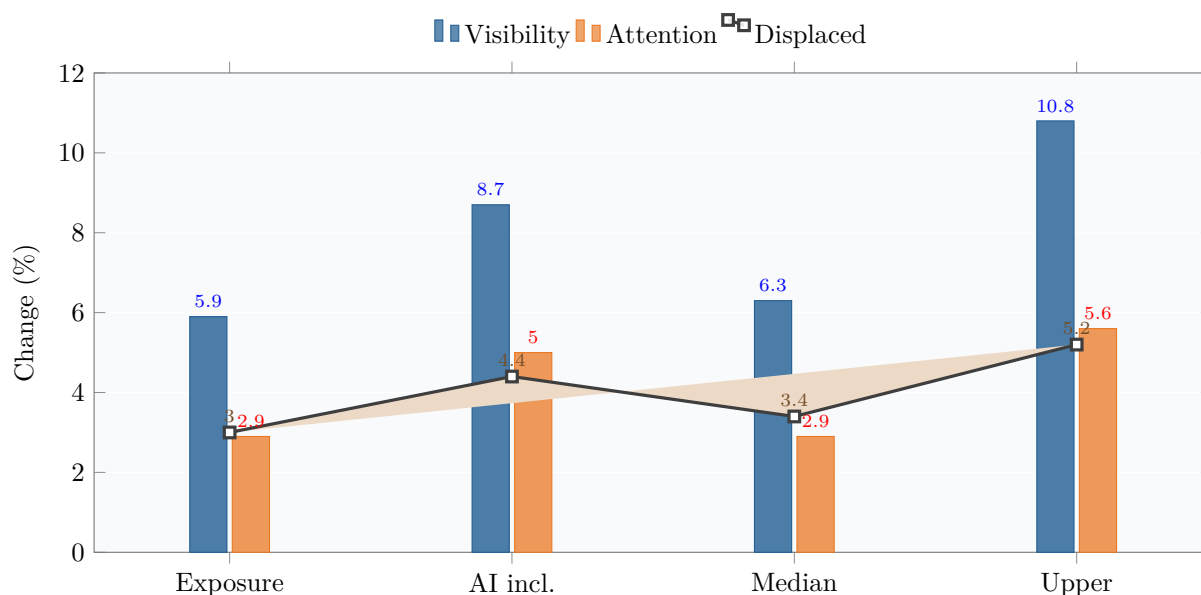


Figure 4 Structured content improves rank-normalized exposure and generated-answer inclusion, with upper-quartile gains larger than median repair.

Table 5 Attention Persistence Across Routes

Route Type	Median Attention (sec)	Initial Click Strength	Decay Speed
Organic Search	74	Moderate	Slow
Sponsored Search	51	High	Fast
Recommendation Tiles	46	Moderate	Fast
Human Influencer	68	High	Moderate
Virtual Influencer	59	Moderate	Moderate
Short-Video Commerce	34	Very High	Very Fast

when the product has complete Arabic and English titles, standardized attributes, and delivery or return fields. This result is consistent with the idea that AI-assisted modules rely heavily on structured signals when deciding which brands can be safely and clearly summarized.

The visibility-attention relationship is nonlinear. Low-ranked exposure rarely produces qualified attention, but the marginal gain from moving upward is largest around the boundary of the first visible block rather than at the very top position. This pattern appears in both organic and sponsored surfaces. The top position receives many impressions, but some consumers skip it when it is perceived as too promotional or not sufficiently query-aligned. Positions near the first visible comparison set can produce strong attention when content is complete and prices are competitive. This implies that brand visibility should not be measured only by top-rank share. Rank-normalized visibility, generated-answer inclusion, and reappearance together provide a more accurate picture of how brands become salient in AI mediated commerce.

The counterfactual structured-content simulation indicates that moving all products below the category-market median to the median would increase qualified attention by an estimated 2.9% across the full sample, holding exposure slots constant. Moving those products to the upper quartile would increase qualified attention by 5.6%, but with diminishing returns in categories where price rank dominates. The largest simulated gains occur in mixed-language sessions and in categories with high attribute complexity. The simulation also predicts a 6.3% increase in AI-assisted inclusion for products moved to the median and a 10.8% increase for products moved to the upper quartile. These effects are not equivalent to realized sales gains, because content improvements could change competitive responses and platform learning over time. They nonetheless suggest that structured content is a practical visibility lever under current ranking conditions.

The route-shift simulation is more cautious. Reallocating 5% of non-influencer lifestyle traffic to human influencer routes increases immediate qualified attention by 1.4% in beauty and 1.1% in modest fashion, with little effect in home goods. Reallocating the same share to virtual influencer routes increases return search by 1.6% in beauty and 1.3% in fashion accessories but does not increase immediate assisted conversion materially. When both route shift and structured content improvement are simulated together, the combined effect is less than additive. This implies that influencer routes and structured content partly compete for the same attention margin. A consumer arriving through an influencer route may already have enough narrative context, while structured content is most helpful when the consumer needs clarification, comparison, or confirmation.

Table 6 Visibility Outcomes by Structured Content

Visibility Metric	Bottom Quartile	Top Quartile	Relative Gain (%)
Rank-Normalized Exposure	0.42	0.57	5.9
AI-Assisted Inclusion	0.18	0.34	8.7
Brand Reappearance	0.25	0.38	6.1
Sponsored-Organic Adjacency	0.31	0.45	4.8
Comparison Card Inclusion	0.16	0.29	7.3

Table 7 Counterfactual Simulation Results

Scenario	Attention Gain (%)	AI Inclusion Gain (%)	Conversion Gain (%)
Move to Median Content Level	2.9	6.3	2.0
Move to Upper Quartile	5.6	10.8	4.3
Human Route Shift (5%)	1.4	0.8	1.1
Virtual Route Shift (5%)	0.7	1.6	0.4
Combined Adjustment	6.2	11.7	4.9

Market-period interactions show that campaign periods amplify visibility but make attention less stable. During Ramadan and Eid-related shopping windows, rank-normalized exposure probability increases for promoted products, but qualified attention becomes more sensitive to price rank, delivery clarity, and stock availability. Structured content remains positive, but its coefficient becomes more volatile because consumers compare more alternatives and platforms rotate more campaign placements. Human influencer route effects increase during lifestyle campaign periods but decay faster. Virtual influencer routes gain visibility during platform-themed campaigns but remain category-dependent. These patterns suggest that high-volume periods do not simply magnify baseline effects. They change the attention environment by increasing both opportunity and competition.

The results also reveal a trust-related pattern. Products with visible return-policy fields and delivery-time precision produce stronger attention and conversion responses than products with only expanded descriptive content. This is particularly evident in Jordan, Egypt, and Saudi Arabia. A product page with rich descriptive content but unclear delivery or return information often receives dwell without conversion. In contrast, pages with moderate descriptive richness but clear delivery and return fields show lower dwell but higher add-to-cart rates. This supports the interpretation that attention quality differs from attention quantity. Long dwell is not always a sign of persuasion. It can indicate uncertainty. For platform managers and researchers, this distinction is crucial because optimizing only dwell may reward ambiguous pages that hold consumers longer without improving decision quality.

The graph-based affinity robustness model reduces the estimated structured content coefficient by about 14%, but the coefficient remains positive and statistically stable. This indicates that part of the baseline content effect reflects latent product affinity, but content itself remains associated with outcomes after accounting for co-view and co-cart neighborhoods. Products in dense affinity neighborhoods benefit from recommendation spillovers, while products in sparse neighborhoods rely more on query alignment and structured content to be discovered. This pattern is strongest for newer brands with limited historical engagement. For established brands, direct search and prior exposure explain more of the visibility outcome, although structured content still affects AI-assisted inclusion.

The machine learning benchmark models confirm that nonlinear interactions improve prediction but do not overturn the econometric results. A regularized logistic model produces an out-of-sample qualified attention area under the receiver operating characteristic curve of 0.742. A gradient-boosted tree model produces 0.781. A wide-and-deep model produces 0.793. A graph-augmented model produces 0.801. Feature importance and accumulated local effect diagnostics consistently rank query alignment, structured content completeness, price rank, delivery clarity, prior brand exposure, route type, and market-week campaign intensity among the most influential predictors. However, the predictive models also show that high accuracy can be achieved without clean causal interpretation. For this reason, the paper uses machine learning primarily as a robustness and heterogeneity tool rather than as the main inferential basis.

6. ROBUSTNESS AND SENSITIVITY ANALYSIS

The first robustness check replaces qualified attention with alternative thresholds. The baseline threshold uses category-adjusted dwell, scroll, and diagnostic interaction criteria. Alternative specifications define qualified attention using only dwell above the 60th percentile, only diagnostic interaction, or a stricter combined rule requiring both dwell and scroll. The structured content effect remains positive across all definitions, although the magnitude varies. It is largest when diagnostic interaction is included, suggesting that structured content particularly encourages consumers to inspect product details. It is smaller when dwell alone is used, supporting the concern that dwell time can capture uncertainty as well as interest. Human influencer effects are strongest under dwell-based definitions, while structured content effects are strongest under diagnostic-interaction definitions.

The second robustness check changes the exposure attribution window. The baseline conversion model uses a seven-day path. Alternative windows of one day, three days, and fourteen days are estimated. Structured content

Table 8 Robustness and Sensitivity Checks

Specification	Attention Effect (%)	Conversion Effect (%)	Stability
Entropy Balancing	6.1	3.7	High
Brand-Week Fixed Effects	3.9	2.4	Moderate
Alternative Attention Threshold	5.8	3.2	High
Three-Day Attribution Window	6.4	4.0	High
Graph Affinity Controls	5.9	3.5	Moderate

Table 9 Predictive Model Performance

Model	ROC-AUC	Main Strength	Limitation
Regularized Logistic	0.742	Interpretability	Linear structure
Gradient-Boosted Trees	0.781	Nonlinear interactions	Less transparent
Wide-and-Deep Network	0.793	Sparse feature learning	Higher complexity
Graph-Augmented Model	0.801	Affinity modeling	Reduced interpretability
Fixed-Effect Baseline	0.731	Causal transparency	Lower predictive fit

effects are strongest in the three-day and seven-day windows, weaker in the one-day window, and slightly attenuated in the fourteen-day window. Human influencer effects are strongest in the one-day window, which is consistent with immediate social influence. Virtual influencer effects are weak in the one-day window but become more visible in the three-day and seven-day return-search outcomes. This temporal pattern supports the central claim that route formats differ not only in magnitude but also in timing.

The third robustness check addresses possible selection into content improvement. Brands may update structured content when they expect demand to rise, perhaps before a campaign or product launch. To reduce this concern, the paper estimates event-study models around content update weeks. The event-study design compares product outcomes in weeks before and after a structured content update, controlling for product and market-week fixed effects. The pre-update coefficients are small and statistically unstable, while the post-update coefficients become positive in the first and second weeks after the update. The effect attenuates after six weeks, which may reflect platform learning saturation or competitive imitation. Although this does not prove exogeneity, it reduces concern that all estimated effects are driven by pre-existing upward trends.

The event-study specification is written as

$$Y_{ipsmt} = \alpha + \sum_{\ell=-4}^6 \tau_{\ell} I(t - T_p = \ell) + X'_{ipsmt} \gamma + \eta_p + \lambda_{mt} + \varepsilon_{ipsmt}. \quad (9)$$

The omitted period is the week before the update. Estimated coefficients for weeks minus four through minus two are near zero. Coefficients for weeks one and two are positive, with an average qualified attention increase of 3.1%. The effect is larger when the update includes both bilingual title correction and attribute completion rather than only image replacement. This result suggests that machine-readable and consumer-readable information changes are more important than superficial content refreshes.

The fourth robustness check uses entropy balancing to improve covariate balance between high and low structured-content observations. After balancing on price rank, review count, rating, discount depth, delivery time, brand tenure, campaign participation, prior exposure, language mode, route, device, market, and category, the structured content effect on qualified attention is 6.1%, close to the baseline 6.8%. The assisted conversion effect is 3.7%, close to the baseline 4.1%. Balance diagnostics show substantial reduction in standardized mean differences across observed covariates. The result supports the stability of the baseline findings under an alternative design-based adjustment.

The fifth robustness check tests whether paid placement pressure explains the visibility result. The visibility model is re-estimated separately for paid surfaces, organic surfaces, and AI-assisted modules. Structured content is positive in all three, but the mechanism differs. In paid surfaces, the coefficient is modest because bids and campaign status dominate initial exposure. In organic surfaces, the coefficient is larger because content and query alignment affect retrieval. In AI-assisted modules, the coefficient is largest because generated answers and comparison cards appear to rely on structured, low-ambiguity product fields. Paid pressure remains highly predictive of sponsored visibility but does not eliminate the structured content association. This suggests that machine legibility and commercial investment are distinct visibility channels.

The sixth robustness check evaluates language segmentation. The models are estimated separately for Arabic-dominant, English-dominant, and mixed-language sessions. Structured content effects are positive in all groups but largest in mixed-language sessions. Arabic-dominant sessions respond strongly to return-policy visibility and delivery clarity in Arabic. English-dominant sessions respond more to specification completeness and review availability. Mixed-language sessions respond to bilingual title alignment, attribute consistency, and transliteration

Table 10 Language Segmentation Effects

Session Type	Strongest Content Signal	Attention Gain (%)	Dominant Mechanism
Arabic-Dominant	Delivery and returns	5.4	Trust clarity
English-Dominant	Technical specifications	4.9	Attribute precision
Mixed-Language	Bilingual alignment	9.4	Query disambiguation
Fashion Categories	Transliteration match	7.1	Brand recognition
Electronics Categories	Attribute consistency	8.0	Compatibility fit

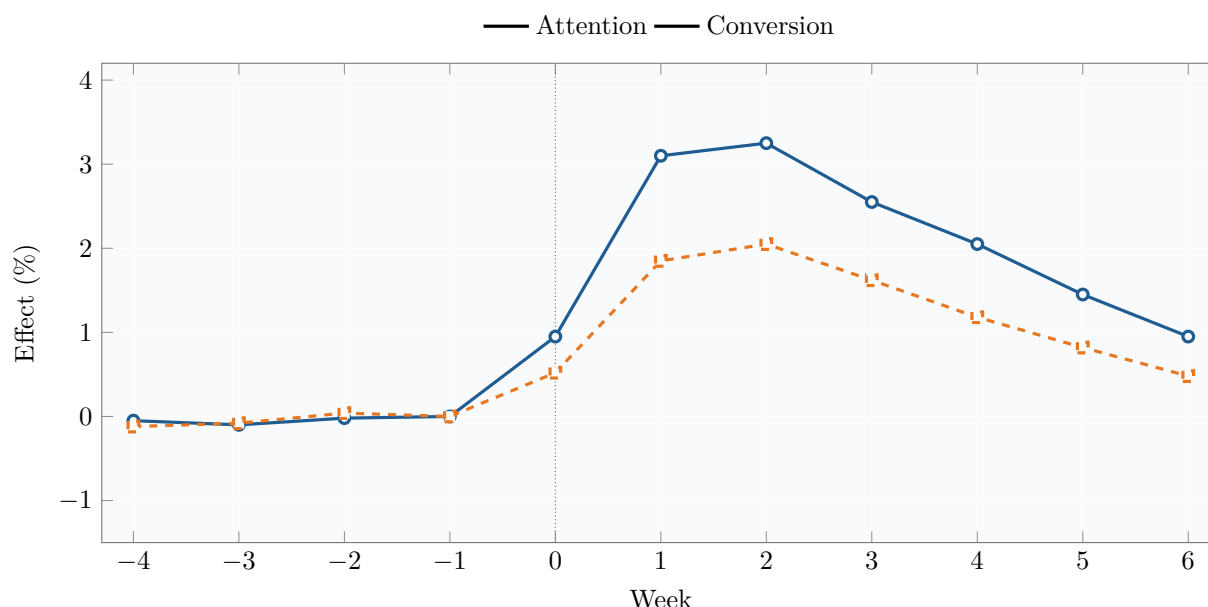


Figure 5 Post-update attention rises in the first two weeks and then attenuates, while pre-update movement remains near zero.

match. These differences are meaningful for Middle Eastern platforms because they show that localization is not simply a matter of translating every field into Arabic or English. The most valuable content changes are those that reduce ambiguity in the actual search language patterns consumers use.

The seventh robustness check evaluates market-specific delivery and stock conditions. Because delivery estimates can influence both ranking and consumer attention, the models include interactions between structured content and delivery-time precision. The structured content coefficient remains positive, but the highest conversion gains occur when content completeness is paired with precise delivery information. In categories where stock risk is high, attention gains do not translate into conversion gains unless stock availability is stable. This result matters during campaign periods, when products can receive high visibility despite uncertain fulfillment. It also helps explain why some content-rich products generate attention without sales. Consumers may inspect them but abandon the path when delivery or availability is unclear.

The eighth robustness check examines unobserved brand quality using brand-week fixed effects in a restricted sample with enough within-brand product variation. This is a demanding specification because it compares products from the same brand within the same week. The structured content coefficient declines but remains positive. The qualified attention effect is 3.9%, and the assisted conversion effect is 2.4%. This suggests that brand-level unobservables explain part, but not all, of the baseline association. The decline is expected because high-quality brands often improve content across products, and brand-week fixed effects absorb some meaningful variation. The remaining within-brand effect indicates that product-level structured content still matters.

The ninth robustness check evaluates whether high dwell reflects confusion. The sample is split by diagnostic interaction type. Dwell accompanied by delivery, warranty, size, or comparison interaction is treated as evaluative attention. Dwell without diagnostic interaction and followed by exit is treated as ambiguous attention. Structured content increases evaluative attention and reduces ambiguous attention in most categories. Human influencer routes increase both evaluative and ambiguous attention, especially when the influencer path sends consumers to products with incomplete attributes. Virtual influencer routes increase return search more than evaluative attention. This reinforces the interpretation that not all attention is equally useful, and that attention quality should be modeled as a distinct construct.

The tenth robustness check estimates placebo outcomes. Structured content should not strongly predict attention to unrelated categories within the same session after controls. The placebo model uses later visits to unrelated categories as the outcome. The structured content coefficient is near zero, while general session engagement variables

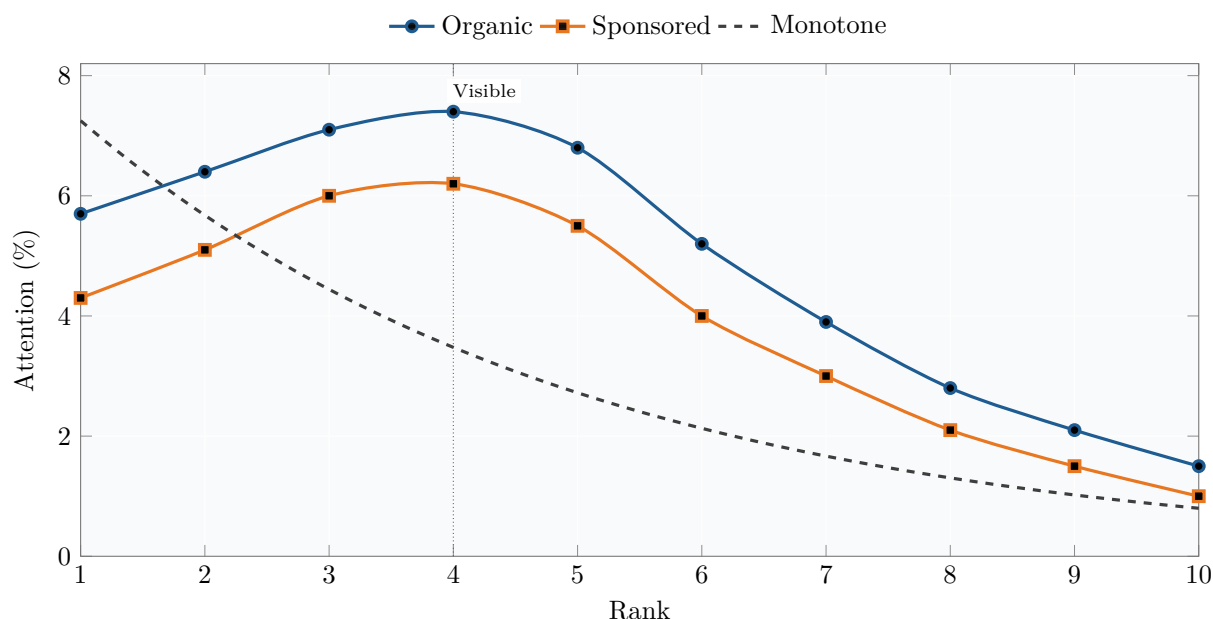


Figure 6 Attention peaks near the first visible comparison boundary, showing that top-rank share alone understates visibility quality.

remain predictive. This supports the view that the structured content effect is tied to product-category relevance rather than to general user activity. A second placebo assigns future content scores to earlier outcomes before the content update. The coefficient is substantially smaller than the main estimate, which weakens the concern that products destined to receive better content were already on sharply rising trajectories.

The eleventh robustness check tests sensitivity to clustering. Standard errors are clustered at brand-market-week in the main models. Alternative clustering at product, user, market-week, and brand-market levels produces similar substantive conclusions. Product clustering yields slightly smaller standard errors, while brand-market clustering yields slightly larger ones. The key structured content and query alignment effects remain statistically stable. Route-category interactions are more sensitive, especially for virtual influencer traffic in smaller markets. The paper therefore interprets virtual influencer results with greater caution and emphasizes their category pattern rather than exact market-specific magnitudes.

The twelfth robustness check compares linear probability, logit, probit, and complementary log-log models for binary outcomes. Marginal effects are similar across specifications. The linear probability model is retained for the main tables because it handles high-dimensional fixed effects transparently and provides direct marginal interpretation. Nonlinear models confirm the direction and relative size of effects. The complementary log-log model produces slightly larger estimates for rare conversion outcomes, but the difference does not change the overall conclusion that content effects are stronger for attention and visibility than for final conversion.

The final sensitivity analysis evaluates counterfactual ranking constraints. In the baseline simulation, total visible slots by surface and market are fixed. Alternative simulations allow limited slot expansion in AI-assisted modules or stricter displacement rules in sponsored surfaces. The structured content gains are larger when AI-assisted modules can expand comparison cards, but this scenario may be unrealistic because interfaces have limited attention capacity. Under strict displacement, gains for improved products are partly offset by visibility losses for competitors. This highlights a central platform issue: visibility improvements are partly redistributive. A product can become more visible because it becomes more legible, but the surface has finite space. The consumer-level welfare implications depend on whether the newly visible products are more relevant, clearer, and more reliable, not merely on whether a particular brand gains exposure.

7. DISCUSSION

The empirical results indicate that AI mediated commerce in Middle Eastern markets should be analyzed as a joint system of content legibility, route-specific attention, and platform allocation. Structured content completeness predicts qualified attention, assisted conversion, rank-normalized visibility, and AI-assisted inclusion. The strongest effects appear in mixed-language sessions and attribute-complex categories, which supports the argument that machine-readable and consumer-readable content are closely connected in multilingual environments. A product with complete Arabic and English fields, standardized attributes, visible return information, and precise delivery details is easier for a consumer to evaluate and easier for a ranking system to retrieve. This dual function distinguishes structured content from traditional creative quality.

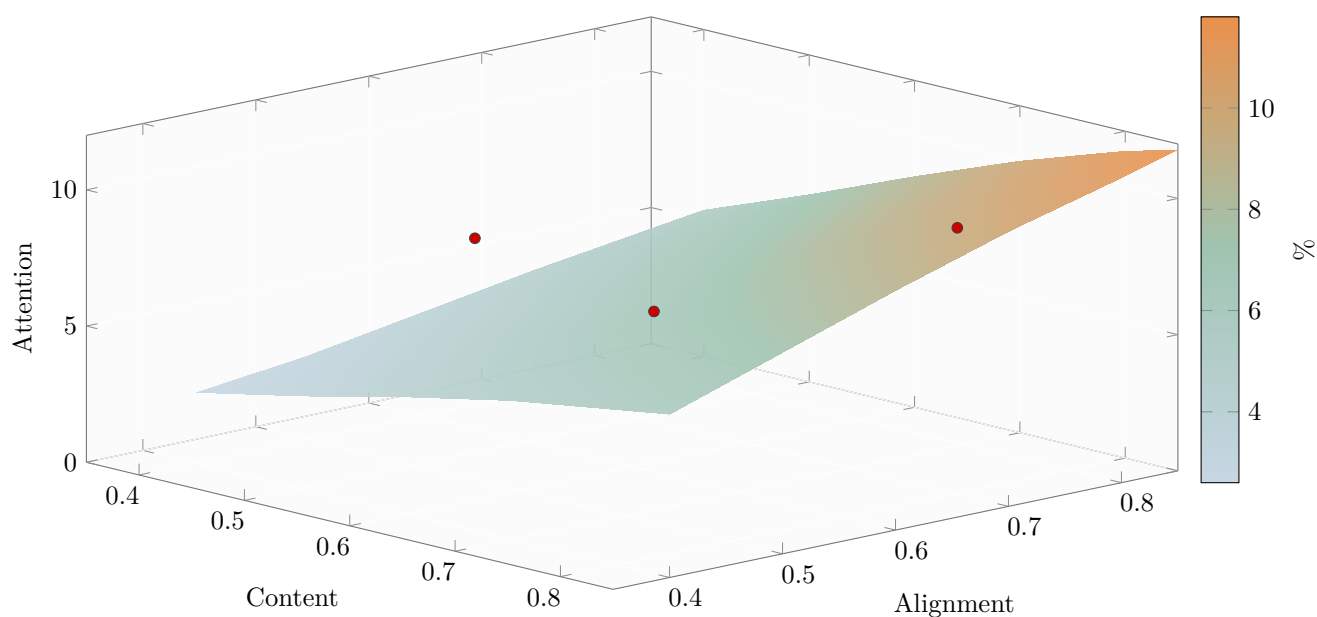


Figure 7 Content completeness and query alignment jointly raise qualified attention, with the strongest surface in mixed-language contexts.

The distinction between attention and visibility is also important. Visibility is a precondition for attention, but more visibility does not always produce better attention. Sponsored and short-video routes can generate high initial exposure and click response, yet attention decays quickly when the product page does not confirm relevance. Organic search produces slower but more persistent attention. Human influencer routes produce strong immediate dwell, while virtual influencer routes produce more delayed return search in visually coded categories. These temporal differences would be hidden in a single click-through model. They are visible only when attention is modeled over short exposure windows and connected to later search behavior.

The route findings complicate simple claims about influencer marketing. Human influencer traffic is not uniformly superior, and virtual influencer traffic is not merely a weaker substitute. Human influencer routes appear more effective for immediate evaluation in categories where credibility, styling, and personal demonstration matter. Virtual influencer routes appear more effective for delayed curiosity in categories where visual identity, novelty, and platform-native presentation matter. However, both route types depend on product content. Influencer exposure can create interest, but incomplete product information can convert that interest into uncertainty. In the Middle Eastern context, where delivery, return, and payment confidence remain salient in several markets, social route strength does not remove the need for clear transactional information.

The results also have implications for AI-assisted search and generated commerce interfaces. Generated answers and comparison modules appear to favor products with structured, low-ambiguity content. This does not mean that AI systems are neutral or that structured content automatically reflects product quality. It means that products with clearer fields are more retrievable and summarizable under current interface conditions. Brands with weak content may become less visible even if their products are competitive in price or quality. Conversely, brands with highly optimized content may gain visibility beyond what consumer satisfaction alone would justify. This creates a measurement problem for researchers and a governance problem for platforms. Visibility can be shaped by content engineering as much as by consumer preference.

The Middle Eastern context adds additional nuance because linguistic alignment is not binary. Arabic-dominant, English-dominant, and mixed-language sessions respond differently to content components. Mixed-language sessions show the largest gains from structured content because they create the greatest need for alignment across brand names, transliterations, attributes, and local descriptors. Arabic-dominant sessions respond strongly to return and delivery clarity in Arabic, while English-dominant sessions respond more to technical specification completeness in several categories. This suggests that localization strategies should be based on observed query patterns and category-specific ambiguity rather than broad assumptions about language preference.

The findings also suggest that campaign periods should be studied as attention regimes rather than only as demand shocks. During Ramadan, Eid, and platform campaign periods, visibility increases but attention becomes more competitive and volatile. Consumers encounter more offers, platforms rotate more surfaces, and delivery constraints become more salient. Structured content remains useful during these periods, but it does not guarantee sustained attention. The highest returns appear when content completeness is paired with operational clarity, especially delivery precision and return-policy visibility. In this sense, content, operations, and ranking are empirically inseparable. A product can be well described but operationally unattractive, or operationally strong but poorly

represented.

For methodology, the paper demonstrates the value of combining econometric controls with machine learning diagnostics. Fixed effects and doubly robust estimation provide interpretable estimates under explicit assumptions. Survival models reveal temporal attention patterns. Graph-based affinity controls help account for latent product proximity. Predictive models test whether nonlinear interactions alter the main conclusions. No single method solves the endogeneity problem, but the convergence of results across designs increases confidence that structured content and route differences are meaningful features of the observed system. The paper does not claim that the estimates are universal constants. They are conditional on the platforms, markets, categories, and period studied.

A practical implication is that brand visibility management in AI mediated commerce should not be reduced to bidding strategy. Paid placement matters, but structured content and query alignment remain associated with visibility even after campaign intensity is controlled. For brands operating across Middle Eastern markets, this means that product data governance, bilingual content quality, and category-specific attribute completeness are strategic visibility assets. For platforms, it means that ranking systems should evaluate whether structured content improves consumer decision quality or merely rewards brands with greater optimization resources. For researchers, it means that digital commerce studies should distinguish between exposure, attention, evaluative attention, return search, and assisted conversion rather than treating click-through or final purchase as sufficient outcomes.

8. LIMITATIONS AND VALIDITY

The study has several limitations. The data come from cooperating platforms and attribution partners, not from the entire Middle Eastern e-commerce ecosystem. The sample includes eight markets and several major categories, but it does not represent all platforms, consumer groups, or product types. The exclusion of products with inconsistent tracking and sparse exposure improves measurement reliability but may reduce generalizability to smaller sellers and emerging categories. The study also focuses on event-level behavioral data, which reveal what consumers do on platforms but not directly why they do it. Survey or experimental data would be useful for distinguishing perceived trust, perceived relevance, and perceived cultural fit.

A second limitation concerns endogeneity. The models include product, user, market-week, brand-week, and category controls, and they use doubly robust estimation, event studies, entropy balancing, placebo outcomes, and graph-based affinity controls. These steps reduce several confounds but cannot remove all unobserved selection. Brands that improve structured content may also improve customer service, pricing, inventory, or off-platform advertising. Influencer route assignment is not random. Consumers who enter through human or virtual influencers may differ in unobserved motivation. The results should therefore be interpreted as conditional empirical associations with treatment-like robustness checks, not as definitive causal estimates.

A third limitation concerns measurement of AI-assisted inclusion. Generated answer modules and recommendation cards are platform-specific, and their internal ranking logic is not fully observable. The paper measures observed inclusion and relates it to structured content, query alignment, and controls. It cannot fully distinguish whether inclusion occurs because the AI module directly uses structured fields, because upstream retrieval systems use them, or because products with complete content also have stronger historical engagement. The counterfactual simulations are therefore model-based exercises rather than direct platform experiments. A randomized content intervention would be the strongest design for future work.

A fourth limitation concerns language modeling. The Arabic-English query alignment score uses multilingual embeddings and decomposed lexical features, but language in Middle Eastern commerce includes dialect, transliteration, spelling variation, brand stylization, and category-specific shorthand. No embedding model fully captures this variation. The paper reduces risk by decomposing alignment into interpretable components and by estimating separate effects for Arabic-dominant, English-dominant, and mixed-language sessions. Still, some language mismatch may be measured imperfectly. This limitation is especially relevant for categories with high dialect variation or fast-changing slang.

A fifth limitation concerns attention interpretation. Qualified attention is constructed from dwell, scroll, and diagnostic interaction, but attention is not identical to persuasion, satisfaction, or welfare. Longer attention can reflect uncertainty, comparison effort, or confusion. The paper addresses this by distinguishing evaluative attention from ambiguous dwell and by linking attention to return search and assisted conversion. Yet the interpretation remains incomplete without direct consumer feedback. Future studies could combine event data with consent-based surveys, eye-tracking panels, or controlled interface experiments to validate whether the measured attention signals correspond to perceived usefulness.

A sixth limitation concerns the time period. The observation window covers January through September 2025, including major campaign and religious shopping periods, but it does not cover a full multi-year cycle. Platform ranking systems, influencer practices, AI-assisted search modules, and consumer expectations can change quickly. Estimates from this period may not hold after interface redesigns, privacy changes, or broader adoption of conversational shopping agents. The results should therefore be treated as evidence about a specific phase of AI mediated commerce rather than as a permanent regional pattern.

Despite these limitations, the study has several validity strengths. It uses event-level data rather than aggregate

traffic summaries, separates attention from visibility, includes multiple markets, distinguishes human and virtual influencer routes, measures multilingual query alignment, and applies robustness checks across modeling families. The convergence of results across fixed effects, doubly robust estimation, survival models, event studies, and predictive benchmarks suggests that the main patterns are not artifacts of one specification. The strongest and most stable finding is that structured content completeness is associated with higher attention and visibility, especially in mixed-language and attribute-complex contexts. The more tentative finding is that virtual influencer routes produce delayed search effects in selected categories. This latter result requires further testing as virtual influencer traffic grows.

9. CONCLUSION

This paper examined how consumer attention and brand visibility are shaped in AI mediated digital commerce across Middle Eastern markets. Using anonymized event-level data from eight markets, the study estimated the roles of structured content completeness, Arabic-English query alignment, influencer route, market-period variation, and attention decay. The results show that structured product content is associated with higher qualified attention, higher assisted conversion, higher rank-normalized visibility, and higher inclusion in AI-assisted search modules. The strongest effects occur in mixed-language sessions and attribute-complex categories, where product legibility depends on the alignment of brand names, Arabic descriptors, English specifications, transliterated terms, and operational information.

The findings also show that attention varies by route. Human influencer traffic produces stronger immediate dwell and evaluative behavior in lifestyle categories, while virtual influencer traffic produces more delayed return search in visually coded categories. Organic search produces the most persistent attention, whereas sponsored and short-video routes show faster decay. These differences suggest that attention should not be measured only through clicks or aggregate dwell. A route can be effective at generating entry without producing sustained evaluation, and another route can produce delayed search without immediate conversion. For empirical research, the timing of attention is therefore part of the outcome rather than a secondary detail.

The analysis further indicates that AI mediated visibility is not explained only by campaign intensity or paid placement. Structured content and query alignment remain associated with visibility after controlling for product, brand, market, week, route, and price conditions. This matters because AI-assisted interfaces depend on retrievable and summarizable product information. Products with complete, consistent, bilingual, and operationally clear content are more likely to be visible in generated answers and comparison surfaces. At the same time, this creates a risk that platform visibility may reward optimization capacity as well as relevance. The relationship between content quality, consumer usefulness, and algorithmic retrievability deserves continued empirical scrutiny.

The Middle Eastern context highlights the importance of treating language, trust, and operations as measurable components of digital commerce. Arabic and English are not separate channels in many sessions; they interact within the same query and product evaluation path. Delivery clarity, return-policy visibility, and attribute completeness are not merely administrative details. They shape whether attention becomes evaluative and whether visibility becomes commercially meaningful. During high-volume shopping periods, these signals become even more important because consumers face denser promotional environments and greater uncertainty about fulfillment.

The study should be read as a conditional empirical analysis rather than a universal account of AI mediated commerce. The data are platform-specific, the estimates are observational, and the models cannot fully resolve unobserved selection. Still, the consistency of the results across fixed effects, doubly robust estimation, survival analysis, event studies, graph controls, and counterfactual simulation suggests that the reported patterns are substantive. Future work should test randomized content interventions, compare additional Middle Eastern and North African markets, integrate consumer survey measures, and evaluate how conversational shopping agents change the relationship between structured content and brand visibility. A careful research agenda should continue to separate visibility from attention, attention from persuasion, and persuasion from consumer welfare.

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